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Numerical and three-factor design investigation for melting process of phase-change spherical capsules in a solar thermal energy storage system

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ABSTRACT

The application of spherical phase-change capsules in solar thermal energy storage systems (STESS) can enhance the sustainability and stability of energy output in solar energy utilization, making it a recent research hotspot. Nonetheless, the influence of neglecting three-dimensional effects through structural simplification on the phase-change process and the criteria for selecting the mushy zone constant in numerical studies are still not well-defined. In this context, a detailed analysis is conducted to investigate the variations in liquid fraction, temperature, flow field, and the time required for complete PCM melting caused by the structural simplification method. Subsequently, a three-factor design analysis is performed on parameters including spherical radius (R), mushy zone constant (A_{mush}) and phase-change temperature range (ΔT) with the aim of establishing a rational basis for selecting the mushy zone constant. Results show that the simplification of the 3D spherical encapsulation structure into a two-dimensional (2D) circular encapsulation structure introduces notable discrepancies in the PCM melting process. In contrast to the 2D circular encapsulation structure, the 3D spherical encapsulation structure demonstrates a reduction in the time required for complete PCM melting by 23 % to 36 %. The influence of simplifying the 3D spherical encapsulation structure into a 2D axisymmetric encapsulation configuration on the numerical simulation of the PCM melting process becomes more pronounced with the increase of R. For the 3D spherical encapsulation structure, the hierarchy of significance regarding the factors influencing the PCM melting process manifests in the sequence: R, A_{mush}, ($R \times A_{mush}$), and ΔT. Furthermore, the formulated correlation based on the aforementioned four factors enables precise prediction of the time required for complete PCM melting within the 3D spherical encapsulation structure and offers a foundation for selecting the appropriate A_{mush} value in numerical simulation. The present work contributes novel insights into the accuracy and rationality of numerical simulations for phase-change processes in the STESS.

1. Introduction

At present, fossil energy sources dominated by coal, oil and natural gas constitute a significant share of the global energy landscape. However, the consumption of these non-renewable energy sources not only triggers energy crises but also imposes severe environmental pollution [1]. As a renewable energy, solar energy is an important alternative to fossil energy because of the advantages of abundant availability, cleanliness and lack of pollution. However, it is affected by factors such as weather, time and geographic location, and suffers from intermittency and volatility problems [2]. Energy storage technologies can play a role in "peak shaving" and "valley filling," and they have been studied and applied in various areas, including solar water heating systems [3] and building energy saving [4]. Latent-heat thermal energy storage technology possesses advantages such as compact heat storage volume, high energy storage density, and a wide temperature range [5], making it a promising approach. Plentiful theoretical, numerical and experimental investigations have been conducted by scholars concerning the phase-change process of substances in latent heat energy storage. Moreover, research in this field is still active [6].

In current numerical studies, to save computational resources and improve calculation speed, several researchers have simplified their models by neglecting three-dimensional effects. They employ structural simplification methods, which involve reducing three-dimensional (3D) structures, such as cubes, cylinders, or spheres, into two-dimensional (2D) structures, such as rectangles, circles, or two-dimensional (2D) axisymmetric encapsulation structures. For example, Younsi and Naji [7] simplified a 3D cuboidal encapsulated structure into a 2D

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rectangular configuration for numerical simulation based on the phase-change process of hydrated salts. The influences of Rayleigh number, aspect ratio, and Prandtl number on the melting process and the solid-liquid interface were investigated. Elbahjaoui and Qarnia [8] simplified the latent heat storage cube unit filled with NEPCM (Al_2O_3/P116) into a 2D rectangular configuration composed of half NEPCM plate and half heat transfer fluid channel for numerical investigations. In addition to the studies that simplify cubes into rectangular enclosures, Seddegh et al. [9] assumed cylindrical shell and tube structures to a rectangular encapsulation structure and investigated the effects of employing pure thermal conduction and combined conduction-convection numerical models on the melting and solidification processes. Abdulateerf et al. [10] ignored the axial heat transfer and simplified the 3D cylindrical enclosure structure into a 2D circular enclosure structure for numerical computations. The scholars mentioned above simplified the cylindrical encapsulation structure into rectangular or circular configurations through assumptions for numerical computations. However, a greater number of scholars have simplified the cylindrical encapsulation structure into 2D axisymmetric configurations for research. Ogoh and Groulx [11] investigated the influence of heat transfer fluid velocity on the thermal storage characteristics of cylindrical latent heat energy storage systems by simplifying the cylinder into a 2D axisymmetric encapsulation structure. Similarly, Pan et al. [12] and Chen et al. [13] also simplified the cylindrical body into a 2D axisymmetric rectangular encapsulation structure when employing numerical models to calculate the phase-change process. In addition to the simplification of 3D cylindrical encapsulation structures into 2D encapsulation structures, some researchers have also numerically investigated computations by simplifying 3D spherical encapsulation structures into 2D circular or axisymmetric encapsulation structures. In the investigation of phase-change packed beds, Dong et al. [14] solely accounted for the flow and heat transfer characteristics in the x and y directions. Similarly, Ding et al. [15] simplified the puck package structure into a 2D circular structure during the numerical investigation. Sattari et al. [16] simplified the 3D spherical encapsulation structure into a 2D axisymmetric encapsulation structure for numerical computations, investigating and analyzing the influences of surface temperature, initial temperature, and spherical radius on the melting process. Hosseinizadeh et al. [17] and Tan et al. [18] also simplified the 3D spherical encapsulation structure into a 2D axisymmetric encapsulation structure in the numerical simulations of phase-change processes. Furthermore, some scholars directly employ 3D models to investigate the external flow and heat transfer processes of spherical structures [19] or the phase-change processes of PCM within encapsulation structures. Tiari and Qiu [20] numerically investigated the influence of finned heat pipes on the thermal storage characteristics of 3D shell and tube energy storage systems. Das et al. [21] conducted an analysis based on a 3D shell-and-tube structure to investigate the influence of adding nanocomposites on the melting process.

It cannot be denied that neglecting the three-dimensional effects of a physical model and simplifying its structure can facilitate the advancement of numerical studies, and in some research, this approach has shown results that are quite consistent with experimental observations. Ben-David et al. [22] conducted a numerical and experimental study to analyze the impact of structural simplification on the phase-change process of gallium in both 2D rectangular and 3D cubic encapsulation structures. They found that the evolution of the solid-liquid interface during the PCM melting process differed significantly between the 2D and 3D encapsulation structures, and the numerical results for the 3D structure were in better agreement with experimental data. However, in the study by Ben-David et al. [22], no quantitative differences in the PCM melting process were specifically highlighted between the 3D cubic and 2D rectangular encapsulation structures. Compared to cubes, spheres are often utilized as encapsulation structures for phase-change materials in particle-packed beds due to their isotropic nature and ease of stacking. They are employed as latent heat energy storage devices to overcome the intermittency and fluctuations in solar energy utilization. Fig. 1(a) and (b) shows a solar water heating system equipped with a spherical phase-change packed bed storage unit. Therefore, it is of significant importance in the field of solar energy to determine whether the 3D effects of spherical structures can be neglected during the phase-change process of PCM, and to quantitatively assess the impact of simplifying 3D spheres on the phase-change heat storage process for spherical packed-bed storage units.

The numerical solution methods for phase-change processes are primarily divided into the strain net and fixed mesh methods. Among these, the fixed mesh method does not require consideration of the temporal and spatial variations of the solid-liquid interface. Instead, it directly determines the position of the solid-liquid interface based on the distribution of the temperature field. This approach has garnered considerable attention. The enthalpy-porosity method based on a fixed mesh [23–25] treats the coexisting solid-liquid region as a porous medium and addresses the natural convection of liquid PCM by introducing source terms in the momentum equation. The enthalpy-porosity method is a commonly used numerical technique for solving solid-liquid phase change problems [26]. In this method, selecting the appropriate A_{mush} has a significant impact on the accuracy of numerical simulations. When the geometric size is 50 × 120 (mm) and the ΔT is 4.7 K, Fadl and Eames [27] investigated the melting process within a cavity filled with Lauric acid and found that employing excessively large A_{mush} values resulted in reduced predicted liquid PCM flow velocities and slower melting rates. Through numerical research of the phase-change process in a cavity with dimensions of 20 × 20 (mm), Hong et al. [28] also discovered that variations in A_{mush} induce alterations in the liquid composition, with a maintained phase-change temperature range of 2 K.

Regarding the selection of the mushy zone constant (A_{mush}), there are currently three main methods: Some scholars analyze the influence of different A_{mush} values on the phase-change process and subsequently choose an optimal value for investigation; Some researchers concurrently conduct numerical simulations and experimental studies, comparing both to determine an appropriate A_{mush} value; Another group of scholars directly adopt values from existing literature. Liao et al. [29] investigated the influence of A_{mush} on the melting rate of n-octadecane in a 2D axisymmetric encapsulation structure (R = 25.415 mm) within the range of $1 \times 10^5\ kg{\cdot}m^{-3}{\cdot}s^{-1} \le A_{mush} \le 1 \times 10^9\ kg{\cdot}m^{-3}{\cdot}s^{-1}$. The PCM used in the study is n-octadecane with a ΔT of 2 K. Results showed that the optimal value of A_{mush} was increased as the Stefan number increased. Similarly employing n-octadecane, Hosseinizadeh et al. [17] contrasted the effects of three distinct A_{mush} values ($1 \times 10^5\ kg{\cdot}m^{-3}{\cdot}s^{-1}$, $1 \times 10^6\ kg{\cdot}m^{-3}{\cdot}s^{-1}$, and $1 \times 10^7\ kg{\cdot}m^{-3}{\cdot}s^{-1}$) on the melting process of PCM within four different-sized (R = 40, 60, 80, 101.66 mm) encapsulation structures. By comparing with experimental results, it was deduced that $1 \times 10^5\ kg{\cdot}m^{-3}{\cdot}s^{-1}$ was the optimal value. Based on the same dimensions (50 × 120 mm) and ΔT (4.7 K), Zhao et al. [30] and Karami and Kamkari [31] separately conducted numerical simulations of the lauric acid melting process within rectangular encapsulation structures, comparing their findings with experimental results. The experimental comparisons between Zhao et al. [30] and Kamkari et al. [32] indicated that $5 \times 10^5\ kg{\cdot}m^{-3}{\cdot}s^{-1}$ was the optimal value. However, Karami and Kamkari [31] demonstrated that $5 \times 10^6\ kg{\cdot}m^{-3}{\cdot}s^{-1}$ was the optimal value by comparing the solid-liquid interface with the experimental result of Kamkari and Groulx [33]. It can be seen that there is no unified value for the most suitable A_{mush} in different studies. Furthermore, Table 1 summarizes some typical studies that employed different methods to select the value of A_{mush}. It should be noted that in geometric size, the size of a round is represented by the radius (R), the size of a rectangle is represented by the length (L) × width (W), the size of a triangle is represented by the length of the side (L), and the size of a cuboid is represented by the length (L) × width (W) × height (H). From existing research, it can be observed that different scholars use different optimal A_{mush} when considering different geometric models and PCMs. Even for the same geometric model and PCM, scholars may obtain

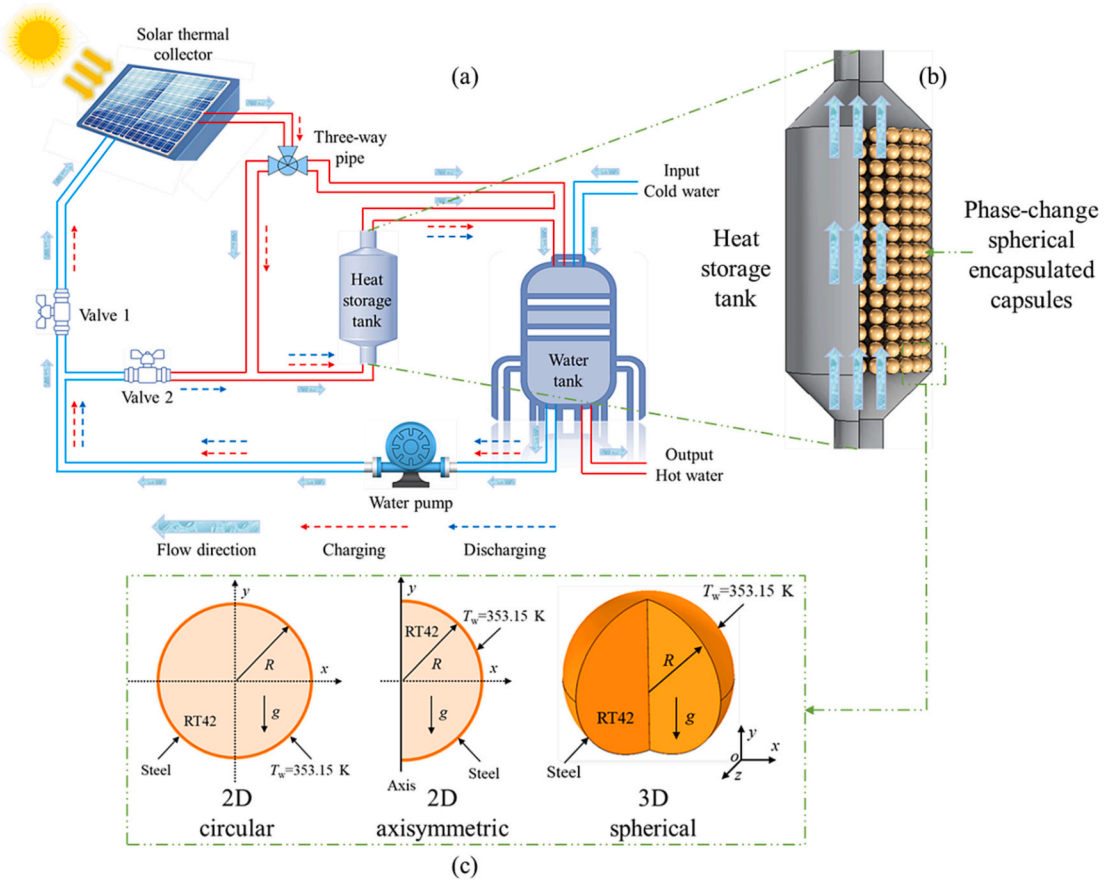


Fig. 1. A solar water heating system with a phase-change thermal storage tank (a), A packed bed of spherical phase-change capsules (b), and simulated cases to investigate the effect of structural simplification on the phase-change process: a 2D circular encapsulation structure, a 2D axisymmetric encapsulation structure and a 3D spherical encapsulation structure (c).

Table 1
The A_{mush} used in some investigations.

References	Mushy zone constant A_{mush} ($kg{\cdot}m^{-3}{\cdot}s^{-1}$)	Phase-change temperature range ΔT (K)	Container geometry	Geometric size (mm)	PCM
Hosseini et al. [17]	1×10^6	0	Round	40, 60, 80, 101.66	Paraffin wax
Al-Abidi et al. [34]	1×10^5	8	Round	75	Paraffin wax
Abdulateef et al. [10]	1×10^9	8	Round	190.5	Paraffin wax
Elsabahy et al. [35]	1×10^6	0	Rectangle	150 × 125	Paraffin wax
Darzi et al. [36]	1×10^5	2	Rectangle	42 × 42, 31.5 × 55, 22 × 80	Paraffin wax
Emam and Ahmed [37]	1×10^6	0	Rectangle	150 × 125	Paraffin wax
Wang et al. [38]	1×10^5	2	Rectangle	22 × 12	Paraffin wax
Nie et al. [39]	1×10^8	2	Rectangle	20 × 20	Paraffin wax
Hong et al. [40]	1×10^8	2	Rectangle	20 × 20	Paraffin wax
Chen et al. [13]	1×10^6	2	Rectangle	80 × 200	Paraffin wax
NematpourKeshteli et al. [41]	1×10^6	8	Cuboid	1000 × 75 × 20	Paraffin wax
Tabassum et al. [42]	1.8×10^{10}	8.7	Triangle	59.5	Paraffin wax
Binet and Lacroix. [43]	1.6×10^6	0	Rectangle	120 × 90	n-octadecane
Jung et al. [44]	1×10^6	1.4	Rectangle	20 × 1565	n-octadecane
Yuan et al. [45]	1×10^5	0	Round	40	Lauric acid
Kamkari and Amlashi [46]	5×10^6	4.7	Rectangle	50 × 120	Lauric acid
Yang et al. [47]	1×10^5	0	Rectangle	5 × 20	Gallium
Brent et al. [25]	1.6×10^6	0	Rectangle	88.9 × 63.5	Gallium
Younsi and Naji [7]	1×10^9	0	Rectangle	25 × 210	Gallium
Pan et al. [12]	7.5×10^5	0	Rectangle	22.58 × 95	$CaCl_2{\cdot}6H_2O$

distinct optimal A_{mush} values through their research. Additionally, it can be observed that the phase-change temperature is not a fixed value but rather exists within a certain range of intervals. This is because commonly used PCM are mostly non-pure substances, where the temperatures of melting and solidification are not equal. For instance, the phase-change temperature range of aluminum alloy is between 919 K

and 930 K [48]. Based on the 2D rectangular encapsulation structure (50 × 120 mm), Kheirabadi and Groulx [49] used numerical methods to investigate the effects of A_{mush} (1×10^3 kg·m^{-3}·s^{-1} ≤ A_{mush} ≤ 1×10^8 kg·m^{-3}·s^{-1}) and ΔT (1 K, 2 K, 5 K) on the phase-change process. The results demonstrated that both A_{mush} and ΔT had an impact on the phase change process. It was particularly noted that the roles of A_{mush} and ΔT in accurately calculating the melting rate were not independent of each other. Therefore, when considering three-dimensional effects, further research and analysis of the influence of A_{mush} and ΔT on the melting rate during the PCM phase change process are meaningful, especially for different radius of spherical encapsulation structures.

Building on the summary of existing research, this study employs the widely used enthalpy-porosity numerical method to investigate the impact of structural simplification on the PCM melting process of the spherical encapsulation structure. Furthermore, a three-factor design approach is utilized to analyze the main and interaction effects of different parameters on the PCM melting process. Based on the spherical encapsulation structure, a comparative analysis of the liquid-phase fraction, temperature distribution, flow characteristics, and the time required for PCM to reach complete melting is conducted. The aim is to investigate the influence of simplifying the 3D spherical encapsulation structure into 2D circular or axisymmetric encapsulation structures on the PCM melting process. Additionally, this study provides a quantitative assessment of the deviations caused by structural simplification methods. Building upon the foundation of the 3D spherical encapsulation structure, a three-factor design methodology is employed to explore the main and interaction effects of the spherical radius (20 mm ≤ R ≤ 50 mm), the mushy zone constant (1×10^5 kg·m^{-3}·s^{-1} ≤ A_{mush} ≤ 1×10^8 kg·m^{-3}·s^{-1}) and the phase-change temperature range (2 K ≤ ΔT ≤ 5 K) on the PCM melting process. Ultimately, a reference basis for the accuracy and rationality selection of A_{mush}.

2. Physical model and computational method

2.1. Physical model

A solar water heating system with a heat storage tank is illustrated in Fig. 1(a). When valve 2 is closed and valve 1 is open, the system undergoes heat storage, during which excess heat is stored in the heat storage tank. When valve 2 is opened and valve 1 is closed, the heat storage unit releases the stored heat. The heat storage tank plays a pivotal role in the entire system. A packed bed of spherical phase-change capsules, used for heat storage, is depicted in Fig. 1(b). Therefore, researching the phase-change process of PCM in spherical capsules is of paramount significance for enhancing the efficiency of the system. As mentioned in the introduction, to analyze the impact of structural simplification on the PCM melting process, three distinct encapsulation structures are established: a 2D circular encapsulation structure, a 2D axisymmetric encapsulation structure and a 3D spherical encapsulation structure, as illustrated in Fig. 1(c). The shell of the spherical container is constructed from steel with a thickness of 2.5 mm. Due to its high thermal conductivity, its thermal resistance impact is neglected in the numerical calculations. The filling phase change material is paraffin RT42, and its thermophysical properties are shown in Table 2 [50]. The influence of the structural simplification method on the PCM melting process is compared and analyzed at different radii (R = 10, 20, 30, 40, 50 mm). Previous investigations have indicated that the mushy zone constant (A_{mush}) influences the intensity of natural convection in liquid PCM, while the phase-change temperature range (ΔT) affects the extent of the non-isothermal mushy zone. However, the extent of significance and interaction effects of A_{mush} and ΔT on the melting process of PCM remains under further investigation. Furthermore, the effects of A_{mush} and ΔT on the phase-change process may vary with different encapsulation structure sizes. Thus, based on the 3D spherical encapsulation structure, a three-factor design approach is employed to investigate the main and interaction effects of spherical radius (20 mm ≤ R ≤ 50 mm), the mushy zone constant (1×10^5 kg·m^{-3}·s^{-1} ≤ A_{mush} ≤ 1×10^8 kg·m^{-3}·s^{-1}), and the phase-change temperature range (2 K ≤ ΔT ≤ 5 K) on the PCM melting process. The three-factor design analysis method will be comprehensively described in Section 2.4. In all numerical computations, the initial temperature (T_{ref}) of the system is maintained at 293.15 K, and the wall temperature of the encapsulation structure is fixed at T_w = 353.15 K.

Table 2
RT42 thermophysical properties.

Properties	Symbols	Value
Thermal conductivity, W·m^{-1}·K^{-1}	k	0.2
Density(solid/liquid), kg·m^{-3}	ρ_s/ρ_l	760/880
Specific heat capacity(solid/liquid), J·kg^{-1}·K^{-1}	$c_{p,s}/c_{p,l}$	1800/2400
Average phase change temperature, K	T_m	317.15
Volume thermal expansion coefficient, K^{-1}	β	0.0001
Latent heat, J/kg	L	141,600

2.2. Governing equations and computational method

In the work, numerical simulations are conducted to investigate the melting process of phase change material (PCM) in three different encapsulation structures. The enthalpy-porosity numerical method [23–25] is used to analyze the solid-liquid phase change process of PCM. In this method, the solid-liquid mushy zone is treated as a pseudo-porous zone, and its "porosity" is represented using the liquid fraction $\gamma(T)$. The Boussinesq approximation is employed to simulate the density variation and thermal buoyancy of the liquid PCM to model the natural convection inside the liquid PCM. The specific assumptions and governing equations used in the study are as follows:

i. The flow of the liquid PCM is transient, laminar, incompressible, and Newtonian flow.
ii. The viscous dissipation and the motion or settling of solid PCM are neglected [51,52].
iii. Density is constant except in the buoyancy term, where the Boussinesq approximation is employed.
iv. The encapsulation capsule shell is made of highly thermally conductive material and thermal resistance is neglected.

Continuity:

$$\rho \nabla \cdot \vec{V} = 0 \tag{1}$$

Momentum:

$$\rho \frac{\partial \vec{V}}{\partial t} + \rho \vec{V} \cdot \nabla \vec{V} = -\nabla p + \mu \nabla \left(\nabla \vec{V} + \left(\nabla \vec{V} \right)^{\mathrm{T}} \right) + \rho g \beta (T - T_m) + S(T) \cdot \vec{V} \tag{2}$$

Energy:

$$\frac{\partial (\rho H)}{\partial t} + \nabla \cdot \left(\rho \vec{V} H \right) = \nabla \cdot (k \nabla T) \tag{3}$$

The equation on enthalpy:

$$\begin{aligned} H &= h + \Delta H \\ h &= h_{ref} + \int_{T_{ref}}^{T} c_p dT \\ \Delta H &= \gamma(T) L \end{aligned} \tag{4}$$

where h is the sensible enthalpy, ΔH is the latent enthalpy, T_{ref} is the initial temperature, L is the latent heat of the PCM, and T is the current temperature.

Based on Darcy's law and the Carman-Kozeny equation, the source term $S(T)$ is introduced in the momentum equation [16,17]:

$$S(T) = -A_{\text{mush}}\frac{(1-\gamma(T))^2}{\gamma(T)^3+\varepsilon} \tag{5}$$

where the A_{mush} is the mushy zone constant. To avoid the situation of being divided by zero when the $\gamma(T) = 0$, the ε is set to 0.001 according to the study of NematpourKeshteli et al. [41]. In the liquid region with the liquid fraction of $\gamma(T) = 1$, the additional source term $S(T)$ is 0, where the momentum equation becomes the equation governing the liquid flow. In the solid region with the liquid fraction of $\gamma(T) = 0$, the $S(T)$ is large to avoid fluid flow. The flow state depends on the mushy zone constant A_{mush} and the liquid fraction when $0 < \gamma(T) < 1$.

$\gamma(T)$ is the liquid fraction in the phase-change process:

$$\gamma(T) = \begin{cases} 0, & T < T_{\text{m}} - \Delta T/2 \\ \dfrac{T-(T_{\text{m}}-\Delta T/2)}{\Delta T}, & T_{\text{m}} - \Delta T/2 < T < T_{\text{m}} + \Delta T/2 \\ 1, & T > T_{\text{m}} + \Delta T/2 \end{cases} \tag{6}$$

In addition, the mathematical form of initial conditions and boundary conditions for the three encapsulation structures should be specified before conducting numerical calculations. As shown in Fig. 1(c), the walls of the three different encapsulation structures maintain a no-slip condition during the computational process. The initial and boundary conditions are shown in Table 3. It should be noted that r represents the distance from the center to the wall surface:

a) For 2D circular and 2D axisymmetric structures: $r = \sqrt{(x^2+y^2)}$;

For 3D spherical structure: $r = \sqrt{(x^2+y^2+z^2)}$.

2.3. Model validations

In the work, the simulations are performed on an Intel Xeon processor E5–2650 (2.30 GHz) with 64 GB of RAM running COMSOL Multiphysics. The separate solver is adopted and the constant Newton iteration method is used to solve the problem. The backward Euler method is used as a time-marching method. The stable solution is obtained when the calculated residual is $<10^{-3}$ in each time step. The grid-independent evaluation is carried out based on a 3D spherical encapsulation structure, where the radius of the spherical structure is 20 mm. The initial temperature of the computational domain is 293.15 K. The surface of the spherical structure is fixed at no-slip condition, and its temperature is set to 353.15 K. It's worth noting that, due to the varying sizes of the encapsulation structures in this work, the maximum grid size is set according to the radius (R) of the encapsulation structure, following the approach described by Wang et al. [53]. The results of the grid independence test are shown in Fig. 2. It can be observed that when the maximum grid size is 0.12 R and 0.10 R, the maximum deviation of the liquid fraction at different time steps is 1.34 %, with an average deviation of 0.95 %. Therefore, it can be considered that the numerical calculations achieve grid independence when the maximum grid size is 0.12 R. For the time step independence test, a maximum time step of 2 s is set because the COMSOL automatically selects small time steps to ensure convergence during calculations [49].

This study validates the accuracy of the numerical method by calculating the melting process of PCM in 2D circular, 2D axisymmetric, and 3D spherical encapsulation structures. The results are compared with the numerical and experimental findings of other researchers. Referring to the study by Al-Abidi et al. [34], a 2D PCM encapsulation structure is constructed. The inner and outer sides of the encapsulation structure serve as the heat transfer fluid, while the middle part is filled with PCM RT82. Comparing the liquid fraction during the PCM melting process obtained in the numerical simulations of this study with the results from the literature [34]. As shown in Fig. 3(a), the predicted trend of liquid fraction closely aligns with the numerical results, with a deviation within 5 %. Referring to the work of Tan et al. [18], a 2D axisymmetric encapsulation structure is constructed. The encapsulation structure has constant temperature boundaries, and the interior is filled with the phase change material n-octadecane. From Fig. 3(b), it can be observed that the numerical and experimental results of Tan et al. [18] exhibit similar trends, but there are still differences numerically. In contrast, the numerical results of this study are closer to the experimental results of Tan et al. [18], with an average deviation within 10 %. Referencing the study by Dong et al. [59], a 3D spherical encapsulation structure is constructed. The outer surface of the encapsulation structure is set as a constant temperature boundary, and the interior is filled with the phase change material n-octadecane. Comparing the liquid fraction during the PCM melting process from the numerical calculations in this study with the experimental results of Dong et al. [59], as shown in Fig. 3(c). The trend of the liquid fraction over time is similar, with an average deviation within 10 %. In summary, the validation against three different encapsulation structures indicates that the numerical model in this study can accurately simulate the PCM melting process.

Table 3
Initial and boundary conditions.

Encapsulation structure	Initial condition	Boundary condition
2D circular	$T = T_{\text{ref}}$, $\vec{u}$=0	r = R: $T = T_{\text{w}}$, $\vec{u}$=0
2D axisymmetric	$T = T_{\text{ref}}$, $\vec{u}$=0	r = R: $T = T_{\text{w}}$, $\vec{u}$=0 x = 0 and – R ≤ y ≤ R: $-\vec{n}\cdot\vec{q} = 0$, $\vec{u}$=0
3D spherical	$T = T_{\text{ref}}$, $\vec{u}$=0	r = R: $T = T_{\text{w}}$, $\vec{u}$=0

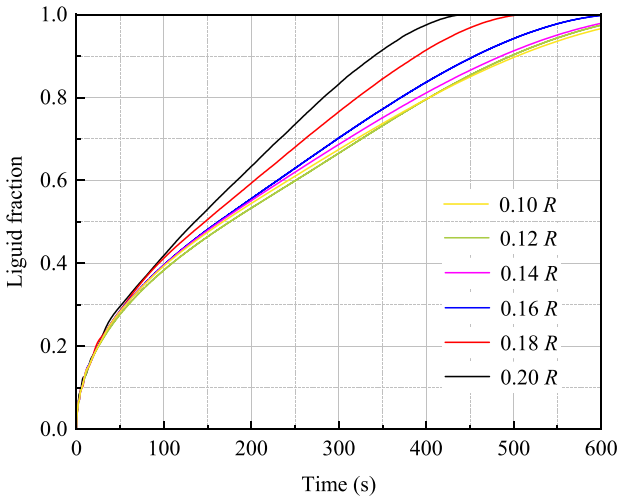


Fig. 2. Variation of liquid fraction with time under different grid configurations.

2.4. Three-factor design method

The design of experimental (DOE) method was initially proposed by the geneticist Ronald Fisher in agricultural experiments and is a technique for studying the relationship between multiple factors and response variables. This method found widespread application in agriculture and the biology sectors [55]. The three-factor design method is a type of experimental design method used for factorial experiments. Its characteristic is the ability to estimate all factor effects, including main effects and interactions on the response variable's significance, while minimizing the number of experiments conducted [56–58]. Jung et al. [44] successfully analyzed the influence of fin number, thermal conductivity, and temperature on the PCM melting process through numerical simulations using a three-factor design. Therefore, to reduce the consumption of numerical computing resources and obtain accurate

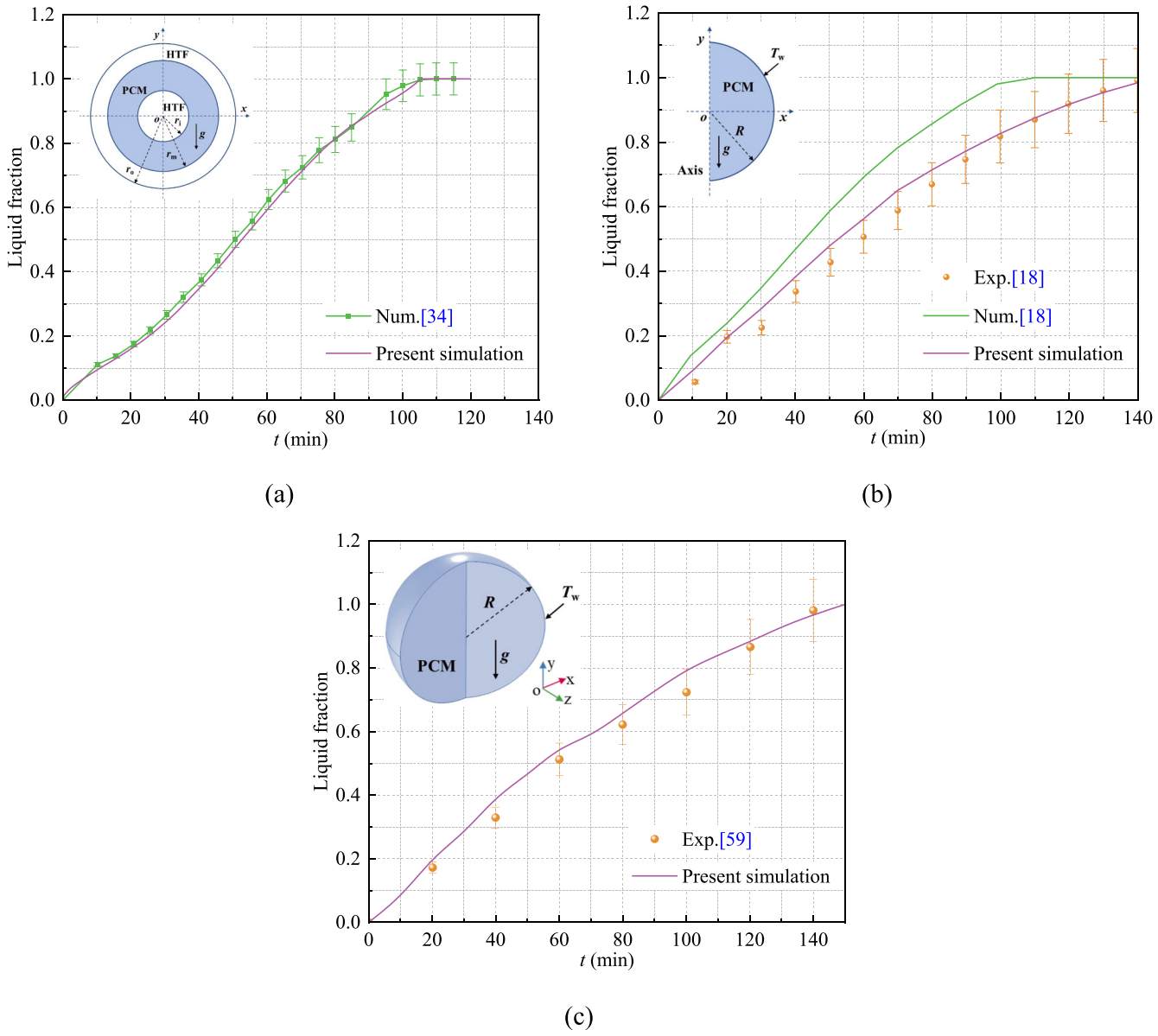


Fig. 3. Comparison of present simulation and literature date in liquid fraction.

analytical results, this work is conducted based on the three-factor design method. The main and interaction effect of spherical radius (20 mm $\leq R \leq$ 50 mm), mushy zone constant (1×10^5 kg·m^{-3}·s^{-1} $\leq A_{\text{mush}} \leq 1 \times 10^8$ kg·m^{-3}·s^{-1}) and phase-change temperature range (2 K $\leq \Delta T \leq$ 5 K) affecting full melting time are analyzed according to the factorial design based on the 3D spherical structure in this work. Table 4 displays eight sets of numerical simulation conditions designed based on the factorial design.

Table 4
Numerical simulation conditions for three-factor design.

Case	R (mm)	A_{mush} (kg·m^{-3}·s^{-1})	ΔT (K)
X_{111}	20	10^5	2
X_{112}	20	10^5	5
X_{121}	20	10^8	2
X_{122}	20	10^8	5
X_{211}	50	10^5	2
X_{212}	50	10^5	5
X_{221}	50	10^8	2
X_{222}	50	10^8	5

Eq. (7) is the data structure equation of the three-factor design used in the calculation. The number of levels specified for R, A_{mush} and ΔT is a, b, and c respectively. The result is x_{ijk} in the case of R_i, $A_{\text{mush}j}$ and ΔT_k. x^{**} represents the population mean of the entire calculation. The $\underline{R}_i$, $\underline{A_{\text{mush}_j}}$ and $\underline{\Delta T}_k$ signifies the effect of factor R_i, $A_{\text{mush}j}$ and ΔT_k. The $\left(\underline{RA_{\text{mush}}}\right)_{ij}$ represents the influence of factor ($R_i \times A_{\text{mush}j}$), which is created by the confounding of R_i and $A_{\text{mush}j}$. Similarly, the $\left(\underline{R\Delta T}\right)_{ik}$ and $\left(\underline{A_{\text{mush}}\Delta T}\right)_{jk}$ stand the effect of factor ($R_i \times \Delta T_k$) and ($A_{\text{mush}j} \times \Delta T_k$), respectively. Furthermore, the $\left(\underline{RA_{\text{mush}}\Delta T}\right)_{ijk}$ signifies an effect of factor of ($R_i \times A_{\text{mush}j} \times \Delta T_k$), which is created through the confounding of R_i, $A_{\text{mush}j}$ and ΔT_k. The $\underline{e}_{ijk}$ represents an effect of factor of ($\underline{E}$). Assuming that the error term follows a normal distribution curve, where σ_E^2 represents the variance.

$$
\begin{aligned}
&x_{ijk} = x^{**} + \underline{R}_i + \underline{A_{mush_j}} + \underline{\Delta T}_k \\
&+\left(\underline{RA_{mush}}\right)_{ij} + \left(\underline{R\Delta T}\right)_{ik} + \left(\underline{A_{mush}\Delta T}\right)_{jk} \\
&+\left(\underline{RA_{mush}\Delta T}\right)_{ijk} + \underline{e}_{ijk} \\
&\underline{e}_{ijk} \sim N\left(0, \sigma_E^2\right) \\
&i = 1, 2, \cdots, a \\
&j = 1, 2, \cdots, b \\
&k = 1, 2, \cdots, c
\end{aligned} \tag{7}
$$

The deviation between the individual data x_{ijk} and total average x^{**} is divided into seven parts, as shown in Eq. (8). The x_i^*, x_j^* and x_k^* are the average of factor R_i, A_{mushj} and ΔT_k, respectively. Similarly, the x_{ij}^*, x_{ik}^* and x_{jk}^* are the average of respective interaction ($R_i \times A_{mushj}$), ($R_i \times \Delta T_k$) and ($A_{mushj} \times \Delta T_k$).

$$
\begin{aligned}
&\left(x_{ijk} - x^{**}\right) = \left(x_i^* - x^{**}\right) + \left(x_j^* - x^{**}\right) + \left(x_k^* - x^{**}\right) \\
&+\left(x_{ij}^* - x_i^* - x_j^* + x^{**}\right) + \left(x_{ik}^* - x_i^* - x_k^* + x^{**}\right) \\
&+\left(x_{jk}^* - x_j^* - x_k^* + x^{**}\right) \\
&+\left(x_{ijk} - x_{ij}^* - x_{ik}^* - x_{jk}^* + x_i^* + x_j^* + x_k^* - x^{**}\right)
\end{aligned} \tag{8}
$$

Eq. (9) is derived by squaring both sides and adding it to i, j and k of Eq. (8). Where the O_T is the total sum of squares and the O_R, $O_{A_{mush}}$ and $O_{\Delta T}$ signify the change of each factor. Besides, $O_{R\times A_{mush}}$, $O_{R\times\Delta T}$ and $O_{A_{mush}\times\Delta T}$ represent variation as a result of interaction. Finally, O_E is an error variation or residual variation.

$$
\begin{cases}
O_{\underline{T}} = O_{\underline{R}} + O_{\underline{A_{mush}}} + O_{\underline{\Delta T}} + O_{\underline{R}\times\underline{A_{mush}}} + O_{\underline{R}\times\underline{\Delta T}} + O_{\underline{A_{mush}}\times\underline{\Delta T}} + O_{\underline{E}} \\
O_{\underline{T}} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\left(x_{ijk} - x^{**}\right)^2 \\
O_{\underline{R}} = \sum_i\sum_j\sum_k\left(x_i^* - x^{**}\right)^2 \\
O_{\underline{A_{mush}}} = \sum_i\sum_j\sum_k\left(x_j^* - x^{**}\right)^2 \\
O_{\underline{\Delta T}} = \sum_i\sum_j\sum_k\left(x_k^* - x^{**}\right)^2 \\
O_{\underline{R}\,\underline{A_{mush}}} = \sum_i\sum_j\sum_k\left(x_{ij}^* - x^{**}\right)^2 \\
O_{\underline{R}\,\underline{\Delta T}} = \sum_i\sum_j\sum_k\left(x_{ik}^* - x^{**}\right)^2 \\
O_{\underline{A_{mush}}\underline{\Delta T}} = \sum_i\sum_j\sum_k\left(x_{jk}^* - x^{**}\right)^2 \\
O_{\underline{R}\times\underline{A_{mush}}} = O_{\underline{R}\,\underline{A_{mush}}} - O_{\underline{R}} - O_{\underline{A_{mush}}} \\
O_{\underline{R}\times\underline{\Delta T}} = O_{\underline{R}\,\underline{\Delta T}} - O_{\underline{R}} - O_{\underline{\Delta T}} \\
O_{\underline{A_{mush}}\times\underline{\Delta T}} = O_{\underline{A_{mush}}\underline{\Delta T}} - O_{\underline{A_{mush}}} - O_{\underline{\Delta T}} \\
O_{\underline{E}} = O_{\underline{T}} - \left(O_{\underline{R}} + O_{\underline{A_{mush}}} + O_{\underline{\Delta T}} + O_{\underline{R}\times\underline{A_{mush}}} + O_{\underline{R}\times\underline{\Delta T}} + O_{\underline{A_{mush}}\times\underline{\Delta T}}\right)
\end{cases} \tag{9}
$$

Eq. (10) shows the respective degree of freedom and Eq. (11) shows the mean square for various factors. Where ϕ represents the degrees of freedom for the factor, and G represents the mean square. Eq. (12) provides the statistical value F_0 and critical value $F(\alpha)$ at the significance level α. The influence of the factor is significant when $F_0 > F(0.05)$ [44].

$$
\begin{cases}
\phi_{\underline{T}} = abc - 1 \\
\phi_{\underline{R}} = a - 1 \\
\phi_{\underline{A_{mush}}} = b - 1 \\
\phi_{\underline{\Delta T}} = c - 1 \\
\phi_{\underline{R}\times\underline{A_{mush}}} = (a-1)\times(b-1) \\
\phi_{\underline{R}\times\underline{\Delta T}} = (b-1)\times(c-1) \\
\phi_{\underline{A_{mush}}\times\underline{\Delta T}} = (a-1)\times(c-1) \\
\phi_{\underline{E}} = (a-1)\times(b-1)\times(c-1)
\end{cases} \tag{10}
$$

$$
\begin{aligned}
&G_\varsigma = O_\varsigma/\phi_\varsigma \\
&\varsigma = \underline{R}, \underline{A_{mush}}, \underline{\Delta T}, \underline{R}\times\underline{A_{mush}}, \underline{R}\times\underline{\Delta T}, \underline{A_{mush}}\times\underline{\Delta T}, \underline{E}
\end{aligned} \tag{11}
$$

$$
\begin{cases}
F_0 = G_\varsigma/G_{\underline{E}} \\
F(\alpha) = F(\phi_\varsigma, \phi_{\underline{E}}; \alpha)
\end{cases} \tag{12}
$$

3. Results and discussion

3.1. The effect of structural simplification

In this section, a detailed analysis is conducted regarding the impact of structural simplification on the PCM melting process. The main focus of the analysis includes the liquid fraction, temperature distribution, and the flow characteristics of the liquid PCM.

Fig. 4 illustrates the variations of liquid fraction and average temperature over time for radii of 10, 20, 30, 40, and 50 mm. The slope of the liquid fraction curve represents the melting rate of the PCM. It can be observed that the liquid fraction in all cases experiences rapid growth with time during the initial stages, followed by a gradual decrease in the slope of the curve until fully melt-down. Exactly, it begins to melt rapidly as the solid PCM comes into contact with the warmer inner wall, leading to a steep increase in the liquid fraction. The temperature gradient between the inner wall and the remaining solid PCM reduces as time progresses. Consequently, the rate of heat transfer decreases, resulting in a slower melting rate as seen in the decreasing slope of the liquid fraction curve. Eventually, the temperature distribution becomes more uniform when the PCM is completely melted.

By comparing the variations of liquid fraction and average temperature over time in the three different encapsulation structures as depicted in Fig. 4, it can be observed that during the initial melting stage, the changes in liquid fraction and average temperature are relatively similar among the three structures. However, as the melting process progresses, the melting rate in the case of the 2D axisymmetric and 3D spherical encapsulation structures becomes notably faster than that in the 2D circular encapsulation structure. The varying melting rates of PCM in different structures lead to differences in the time required for PCM to reach complete melting. As evident from Fig. 5, at the same value of R, the time required for complete melting of PCM is significantly shorter in the case of the 2D axisymmetric and 3D spherical encapsulation structures compared to the 2D circular encapsulation structure. To quantitatively illustrate the differences in the PCM melting process among different structures, Fig. 6 presents the variations in the time required for complete melting of PCM in different encapsulation structures, based on Eq. (13). Where t_c(R) is the time required for PCM to reach complete melting in the 2D circular encapsulation structure at radius R, and $t_s(R)$ is the time required for PCM to reach complete melting in the 2D axisymmetric or 3D spherical encapsulation structure at the same radius. It can be observed that within the scope of the study, compared to the 2D circular encapsulation structure, the reduction in the time required for PCM to achieve complete melting is in the range of 16 % to 36 % for the 2D axisymmetric and 3D spherical encapsulation structures. However, the relationship between the reduction in $N(t)$ and the R of the encapsulation structure is not monotonic. This is because there are many factors that contribute to the differences in the PCM melting process. One of the reasons for the significantly faster PCM melting process in the 2D axisymmetric and 3D spherical encapsulation

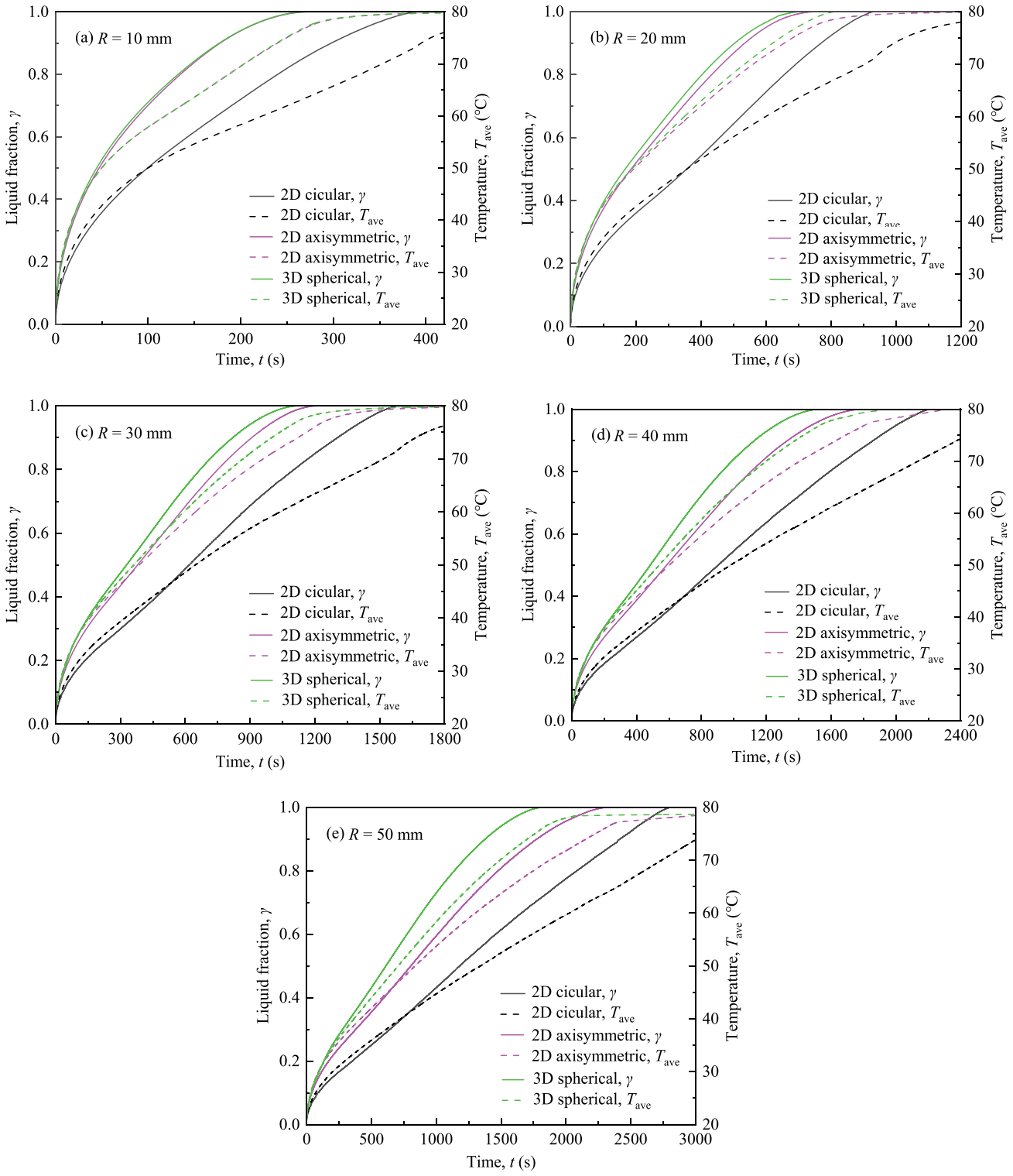


Fig. 4. Variations of liquid fraction and average temperature during the melting process with different spherical radii. (a) $R = 10$ mm (b) $R = 20$ mm (c) $R = 30$ mm (d) $R = 40$ mm (e) $R = 50$ mm.

structures compared to the 2D circular encapsulation structure is the difference in the heat transfer surface area per unit volume. For the 2D circular encapsulation structure, the heating surface can be understood as the circumference, which is $2\pi R$. The melted region undergoing heating is the internal area, which is πR^2. Therefore, the heat surface area per unit volume is $2/R$ for the 2D circular encapsulation structure. The 2D axisymmetric encapsulation structure has the same heat surface area per unit volume as the 3D spherical encapsulation structure due to its rotational characteristics. For the 3D spherical encapsulation structure, the heating surface is the surface area of the spherical structure, which is $4\pi R^2$. The melted region is the volume of the spherical structure, which is $4/3\ \pi R^3$. Therefore, the heat surface area per unit volume is $3/R$ for the 3D circular encapsulation structure. Similarly, the heat surface area per unit volume the 2D axisymmetric encapsulation structure is also $3/R$. For the same R, the heat transfer surface area per unit volume in the 2D axisymmetric and 3D spherical encapsulation

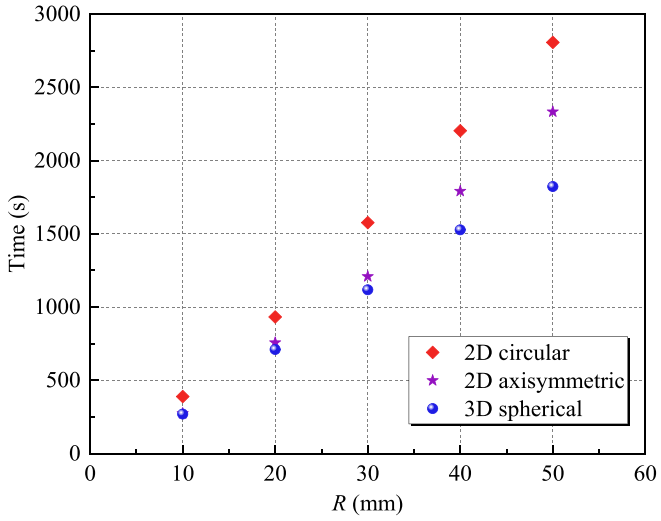


Fig. 5. The time for complete melting of PCM in different structures as a function of *R*.

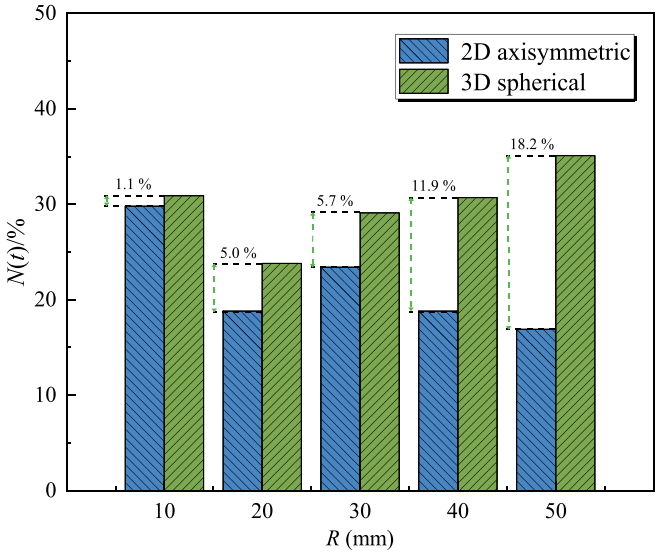


Fig. 6. Differences in the time for complete melting of PCM in different structures.

structures is 1.5 times that of the 2D circular encapsulation structure. Therefore, simplifying the 3D spherical encapsulation structure to a 2D circular encapsulation structure reduces the heat transfer surface area per unit volume, slows down the melting rate, and consequently introduces noticeable discrepancies in the numerical solution of the PCM melting process. As shown in Fig. 4, at the same time, the liquid fraction and average temperature in the smaller-sized encapsulation structures are higher compared to those in the larger-sized encapsulation structures. Furthermore, as observed from Fig. 5, with the increase of *R*, the time required for complete melting of PCM in all three encapsulation structures progressively increases. This can also be explained using the heat exchange surface area per unit volume within the encapsulation structure. For the 2D circular encapsulation structure, the surface area per unit volume is $2/R$, while for the 2D axisymmetric and 3D spherical encapsulation structures, it is $3/R$. It can be observed that for smaller *R* values, the heat exchange area per unit volume between PCM and the inner wall is larger in all three encapsulation structures. Therefore, when *R* is smaller, the heating can occur at a faster rate, resulting in a quicker melting rate of PCM.

$$N(t) = \frac{t_c(R) - t_s(R)}{t_c(R)} \times 100\% \tag{13}$$

Furthermore, in Fig. 4(a), it can be observed that the trends of liquid fraction and average temperature with the progression of the melting process are quite similar in the 2D axisymmetric and 3D spherical encapsulation structures. Similarly, in Fig. 6, it can be observed that the difference in $N(t)$ between the 2D axisymmetric and 3D spherical encapsulation structures is only 1.1 % when $R = 10$ mm. This indicates that for smaller encapsulation shell radii ($R = 10$ mm), assuming the 3D spherical encapsulation structure as a 2D axisymmetric structure results in relatively minor computational discrepancies. However, from Fig. 4 (b) to Fig. 4(e), it is evident that with an increase in *R*, the melting rate of PCM in the 3D spherical encapsulation structure is notably faster than in the 2D axisymmetric structure. Additionally, the magnitude of enhancement in the melting rate tends to increase as *R* increases. Similarly, in Fig. 5, it can also be observed that the time required for complete melting of PCM in the 3D spherical encapsulation structure is less than that in the 2D axisymmetric encapsulation structure. From Fig. 6, it can be seen that with the increase in *R*, the difference in $N(t)$ between the 2D axisymmetric and 3D spherical encapsulation structures increases from 1.1 % to 18.2 %. This indicates that when the radius of the encapsulation structure is large, using the simplification method of the 2D axisymmetric structure may introduce significant errors in the numerical simulation.

To further analyze the distinctions in the melting process of PCM among different encapsulation structures, particularly the reasons behind the variations in the melting process between the 2D axisymmetric and 3D spherical encapsulation structures, an analysis of the distribution of liquid fraction, temperature, and liquid PCM flow characteristics in the encapsulation structures with $R = 10$ mm and 50 mm is conducted. Fig. 7 illustrates the evolution of the liquid fraction contour and temperature distribution over time at $R = 10$ mm. Fig. 8 depicts the natural convection characteristics of the liquid PCM formed after the heating-induced melting process within the encapsulation structure at $R = 10$ mm, resulting from buoyancy and density changes of the PCM. It is worth noting that different plotting methods are used for the analysis of the three encapsulation structures to distinguish different structures. For the 2D circular encapsulation structure, only the *xoy* plane is shown in Fig. 7(a) and Fig. 8(a). For the 2D axisymmetric encapsulation structure, in addition to the *xoy* plane, the three-dimensional shapes in Figs. 7(b) and 8(b) are obtained by rotating the 2D axisymmetric structure. For the 3D spherical encapsulation structure, different cutting planes are taken for analysis, as illustrated in Figs. 7(c) and 8(c). A thin liquid layer appears near the heated wall during the initial stage of melting. Meanwhile, there is a mushy zone between the solid PCM and the liquid layer. From Fig. 7, it can be observed that the liquid fraction and temperature distribution in the 2D circular encapsulation structure exhibit characteristics of rotational symmetry at 120 s. Additionally, as evident from Fig. 8(a), the velocity of the liquid PCM at this moment is relatively low, and the natural convection intensity is weak. This indicates that at this point, the PCM in the 2D circular encapsulation structure is still in a heat conduction-dominated heat transfer stage. However, at 120 s, evident natural convection motion is already present in both the 2D axisymmetric and 3D spherical encapsulation structures. This motion displays a higher concentration of liquid PCM and high-temperature regions in the upper portion compared to the bottom. This indicates that the natural convection motion, driven by buoyancy and density variations, has begun to exert a significant influence on the melting process, which cannot be ignored. As time progresses, the liquid PCM moves upwards, transferring more heat to the solid PCM in the upper part. This is similar to the phenomenon observed by Tan et al. [18] during the experimental visualization of the PCM melting process. At 180 s and 240 s, the maximum natural convection velocities in the three different encapsulation structures are as follows: 7.36×10^{-4} m/s, 7.76×10^{-4} m/s and

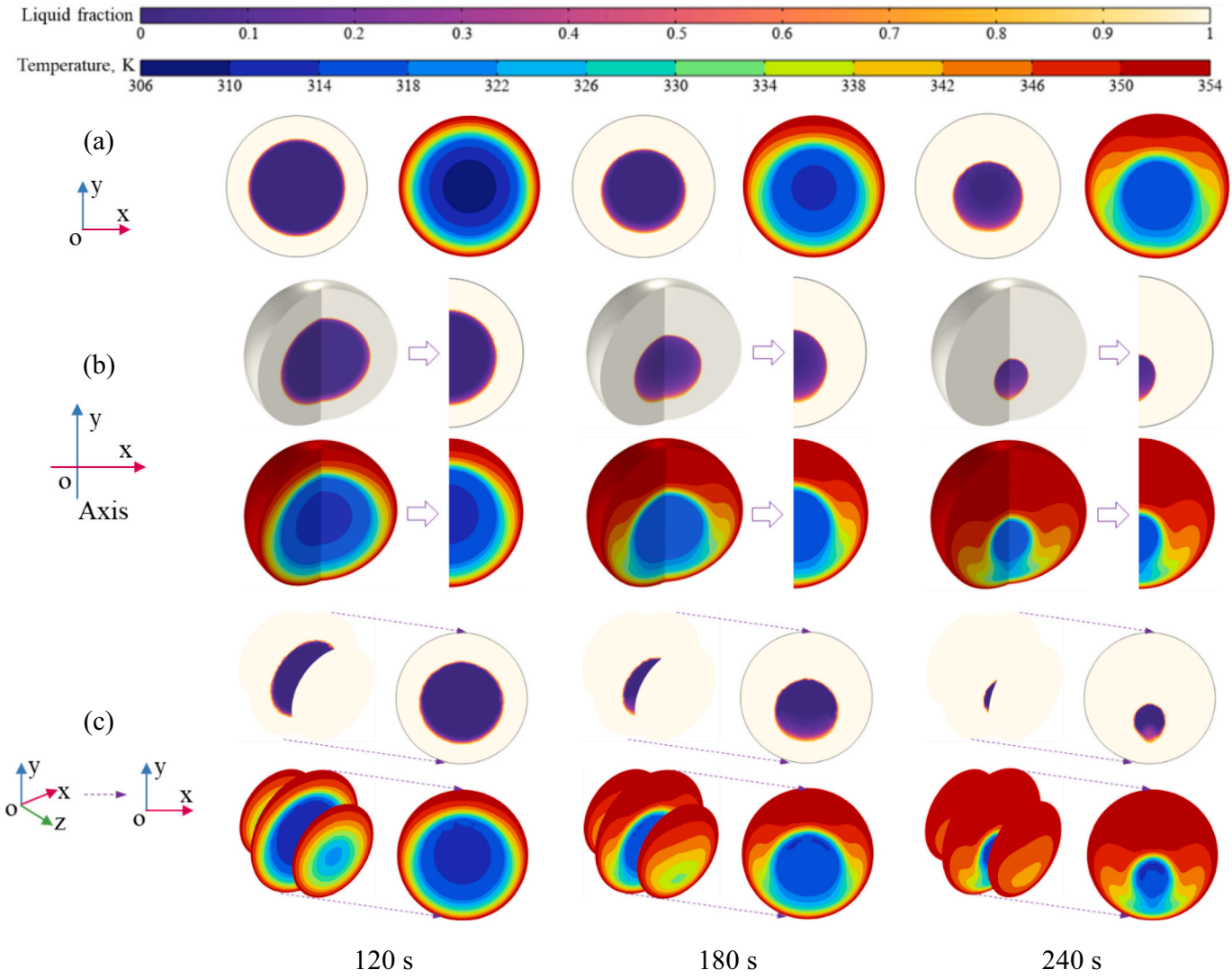


Fig. 7. The evolution process of liquid fraction and temperature distribution over time when $R = 10$ mm. (a) 2D circular model (b) 2D axisymmetric model (c) 3D spherical model.

7.64×10^{-4} m/s. The increased intensity of natural convection in the liquid PCM results in a notably faster melting rate of the PCM in the upper regions of all three encapsulation structures compared to the lower regions.

Furthermore, in Figs. 7 and 8, it can be observed that in both the 2D axisymmetric and 3D spherical encapsulation structures, at different time intervals, the contours of liquid fraction, temperature distribution, and flow characteristics exhibit similar features. This corroborates the observations in Figs. 4(a) and 5, where the trends of liquid fraction and average temperature change over time, as well as the time required for complete melting of PCM, are essentially consistent for both structures when $R = 10$ mm.

Adopting the same plotting approach as when $R = 10$ mm, Fig. 9 provides the contour plots of liquid fraction and the evolution of temperature distribution over time for the three different encapsulation structures when $R = 50$ mm. Fig. 10 illustrates the natural convection characteristics of liquid PCM at different time steps within the encapsulation structures. Firstly, it is evident that the liquid fraction and temperature distribution of PCM already exhibit disparities at 600 s, and these differences gradually become more pronounced as time progresses. At 600 s, the bottom of all encapsulation structures exhibits Rayleigh-Bénard flow characteristics. Similar phenomena have been observed in the study by Nie et al. [39]. This type of flow accelerates the melting rate at the bottom. As time progresses, the Bénard convection continually forms, merges, and evolves, influencing the melting process of solid PCM at the bottom of the encapsulation structures. Similar to $R = 10$ mm, the formation of clockwise and counterclockwise vortices due to the flow of liquid PCM along the inner wall also occurs at $R = 50$ mm. The convective flow patterns created by these vortices play an important role in accelerating the melting process in the upper regions. This results in a higher melting rate of PCM in the upper part of the encapsulation structures compared to the bottom. Similarly, as can be seen in Fig. 10, it can be observed that there are significant differences in velocity distribution among the three different structures. The velocity distribution decreases in the following order: 3D spherical, 2D axisymmetric and 2D circular encapsulation structure. The differences in the strength of natural convection result in varying melting speeds. Therefore, the melting speed of PCM from fastest to slowest is as follows: 3D spherical, 2D axisymmetric and 2D circular encapsulation structure.

In Figs. 7 and 8, it can be observed that the liquid fraction distribution, temperature distribution, and flow field distribution of PCM in the three different encapsulation structures all exhibit characteristics of symmetry when $R = 10$ mm. Particularly, the PCM melting process in the 2D axisymmetric structure and 3D spherical structure exhibits strong similarities. However, a notable difference is seen in Figs. 9 and 10, where only the 2D axisymmetric encapsulation structure continues to

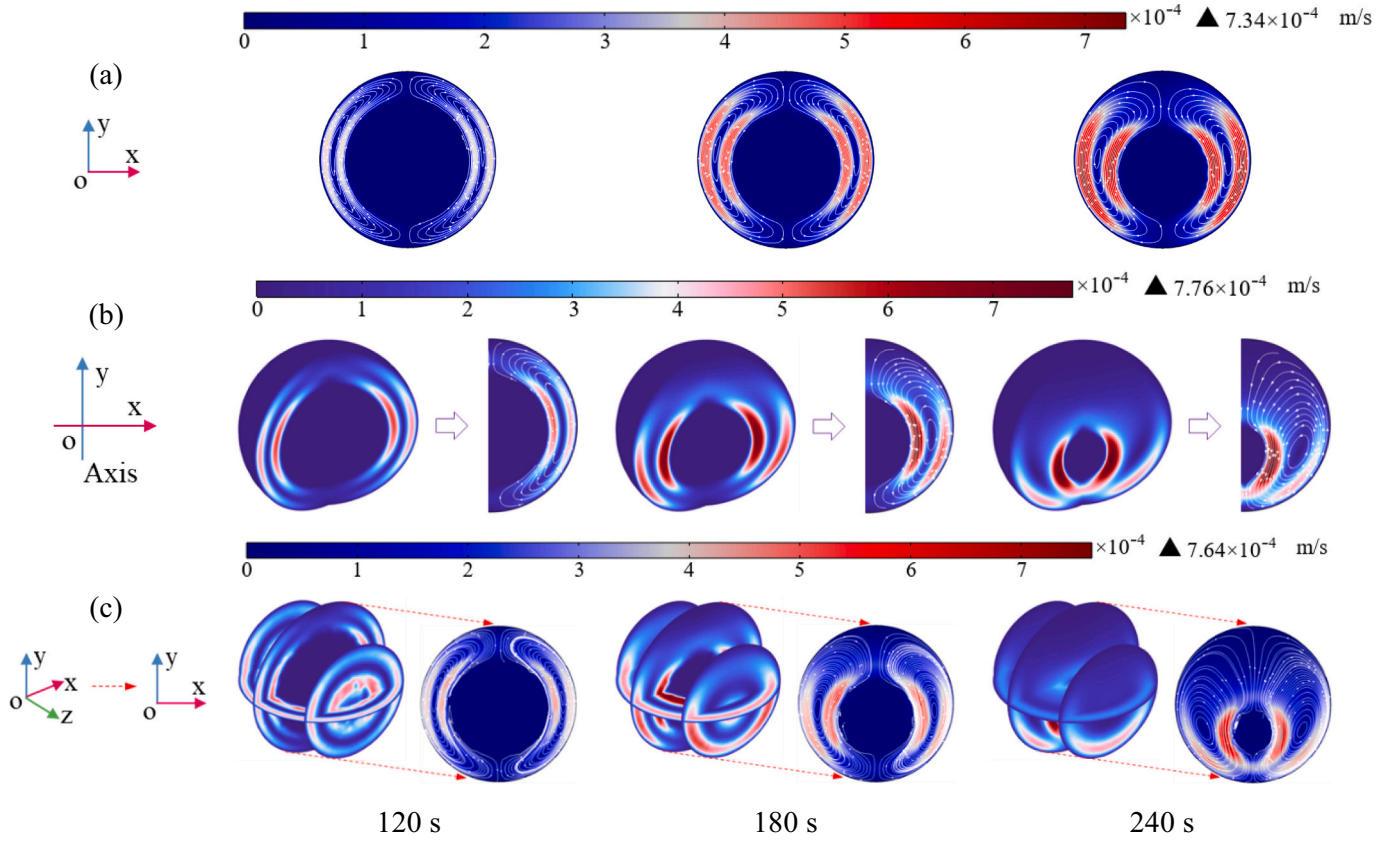


Fig. 8. The evolution process of flow fields over time when $R = 10$ mm. (a) 2D circular model (b) 2D axisymmetric model (c) 3D spherical model.

maintain characteristics of symmetry. This is due to the fact that when the 3D spherical structure is simplified into a 2D axisymmetric structure, the resulting structure has symmetry along any straight line passing through the center of the sphere. Nevertheless, as the size increases, enhanced natural convection disrupts the complete symmetry of the flow field during the PCM melting process. Consequently, the differences in the melting process of PCM between the 2D axisymmetric and 3D spherical encapsulation structures become more pronounced as the radius increases. To further analyze the differences in the melting processes between the 2D axisymmetric structure and 3D spherical structure, an analysis of Fig. 8(b) and (c), as well as Fig. 10(b) and (c), is conducted. Both at $R = 10$ and 50 mm, it can be observed that in the 3D spherical structure, the liquid PCM forms small Bénard cells in different directions, which lead to flow patterns characterized by randomness and volatility. The stochastic and fluctuating flow characteristics enhance the natural convective heat transfer process of the liquid PCM within the encapsulation structure. Compared to the 3D spherical encapsulation structure, the axisymmetric flow pattern assumes uniform symmetry in the flow based on geometric simplification, thereby overlooking the randomness and fluctuations in the flow. This is the reason for the differences in the PCM melting process between the 2D axisymmetric and 3D spherical encapsulation structures. The maximum velocities in the two structures are 7.76×10^{-4} m/s and 7.64×10^{-4} m/s when $R = 10$ mm. It indicates that the complexity of the flow field caused by Bénard cells has a minor impact during the melting process at smaller radii. However, the maximum velocities are 2.73×10^{-3} m/s and 3.61×10^{-3} m/s when $R = 50$ mm, which show significant differences. Moreover, the maximum velocities in the 3D spherical structure are consistently higher than those in the 2D axisymmetric structure at different times, indicating a significant increase in the strength of natural convection in the 3D spherical structure. In contrast to the 2D axisymmetric encapsulation structure, the heightened intensity of natural convection in the liquid-phase PCM engenders a reduction in the time required for complete melting of PCM in the 3D spherical encapsulation structure by 509.7 s. Therefore, the method of simplifying the 3D spherical structure to a 2D axisymmetric structure results in acceptable errors when the characteristic scale is small. However, for larger values of R, due to the inherent non-uniformity and stochastic nature of flow within 3D space, the simplification of the 3D spherical encapsulation structure into a 2D axisymmetric encapsulation structure significantly impacts the numerical computation results.

In conclusion, simplifying the 3D spherical encapsulation structure into a 2D circular encapsulation structure results in a reduction of the heat transfer area per unit volume and neglects flow characteristics in other directions when employing numerical methods to solve the melting process of PCM within a spherical encapsulation structure. This simplification can significantly impact the computed results of the melting process. Simplifying the 3D spherical encapsulation structure into a 2D axisymmetric encapsulation structure leads to an increase in the numerical computation error of the PCM melting process as R increases. Therefore, it is imperative to thoroughly account for the actual geometric characteristics of the spherical encapsulation structure and the 3D nature of flow during the PCM melting process to attain accurate numerical computation results.

3.2. Analysis with three-factor design method

In this section, the three-factor design analysis method is employed to process and analyze the numerical calculation results based on the conditions in Table 4. Fig. 11 illustrates the temperature distribution and streamline patterns at 300 s for the eight different conditions. At 300 s, for the structure with a radius (R) of 20 mm, a larger portion of the enclosed region experiences natural convection and contains liquid PCM compared to the spherical encapsulation structure when $R = 50$ mm.

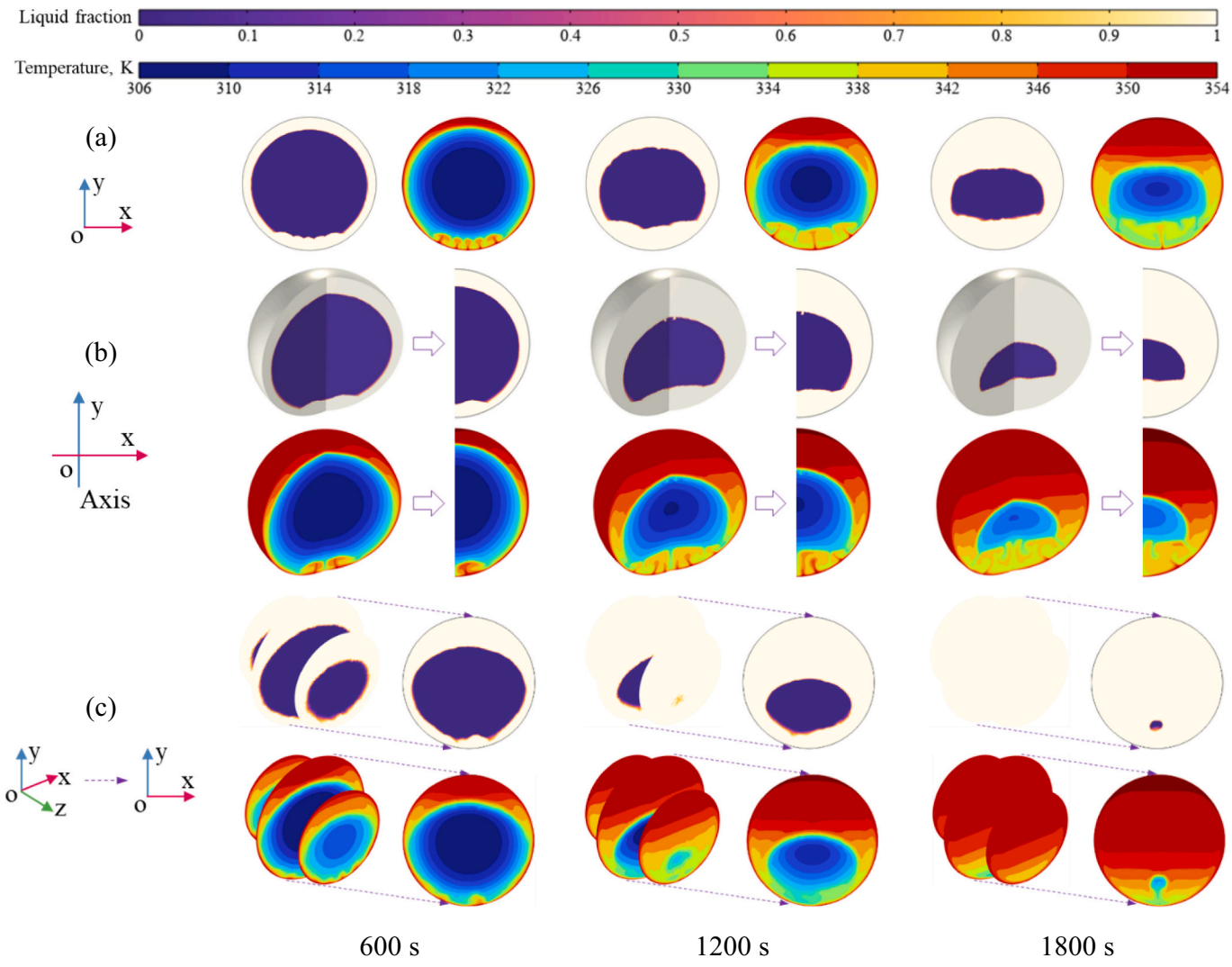


Fig. 9. The evolution process of liquid fraction and temperature distribution over time when $R = 50$ mm. (a) 2D circular model (b) 2D axisymmetric model (c) 3D spherical model.

This indicates that the R significantly influences the melting process of the PCM. A larger proportion of liquid PCM with higher temperatures is present when the mushy zone constant (A_{mush}) is smaller. This effect can be attributed to the reduction in A_{mush}, leading to a decrease in the source term $S(T)$ in Eq. (5). Consequently, the resistance source term in Eq. (2) is diminished, resulting in an overall enhancement of the natural convection strength and melting rate of the liquid PCM within the encapsulation structure. The magnitude of the phase-change temperature range (ΔT) does not significantly influence the extent of non-isothermal slurry region or the strength of natural convection within the slurry region.

Fig. 12 shows the variation of liquid fraction with time during the melting process for different operating conditions. The time required for PCM to reach full melt-down in the spherical encapsulation structure is shorter when the R is smaller. Meanwhile, the influence of the A_{mush} and ΔT on the melting process is relatively small. With the increase in R, the time required for PCM to achieve complete melting significantly increase due to the heat exchange surface area per unit volume decreases. Besides, the influence of A_{mush} and ΔT on the melting rate becomes more pronounced. As the R increases, the flow and heat transfer processes of the liquid PCM become more intricate, with both A_{mush} and ΔT exerting a heightened impact on the melting behavior. Comparatively, the influence of A_{mush} is more substantial, followed by the influence of ΔT.

Furthermore, by taking the time required for PCM to reach complete melting ($\gamma = 1$) as the dependent variable, a quantitative study is conducted to investigate the influence of three factors: spherical radius (R), mushy zone constant (A_{mush}), and phase-change temperature range (ΔT) on the melting process. Table 5 provides the time required for PCM to reach complete melting under different operating conditions. The standardized effects of different factors obtained from the factorial analysis are depicted in the normal plot shown in Fig. 13. It can be observed that the most significant factors influencing the melting process are the R and the A_{mush}. Other factors have relatively minor effects on the time required to achieve complete melting. Therefore, the factors with insignificant effects are combined into the error term ($\underline{E}$), resulting in a new analysis of the variance table as shown in Table 6. It should be noted that factor ΔT is retained following the principle of not combining main effects. Furthermore, the interaction factor between the R and the A_{mush} is taken into consideration due to the significant main effects of R and A_{mush}. In Table 6, O represents the sum of squares, ϕ represents the degrees of freedom. The P is a value obtained through a significance test method. In statistics, when $P < 0.05$, an influence is considered to be present [54]. F_0 and P_0 are the F-value and P-value corresponding to each factor. The F-values and P-values corresponding to different factors are presented in Table 5. From the values of F_0 and P_0 in Table 6, it can be observed that all four factors have an impact on the time required for

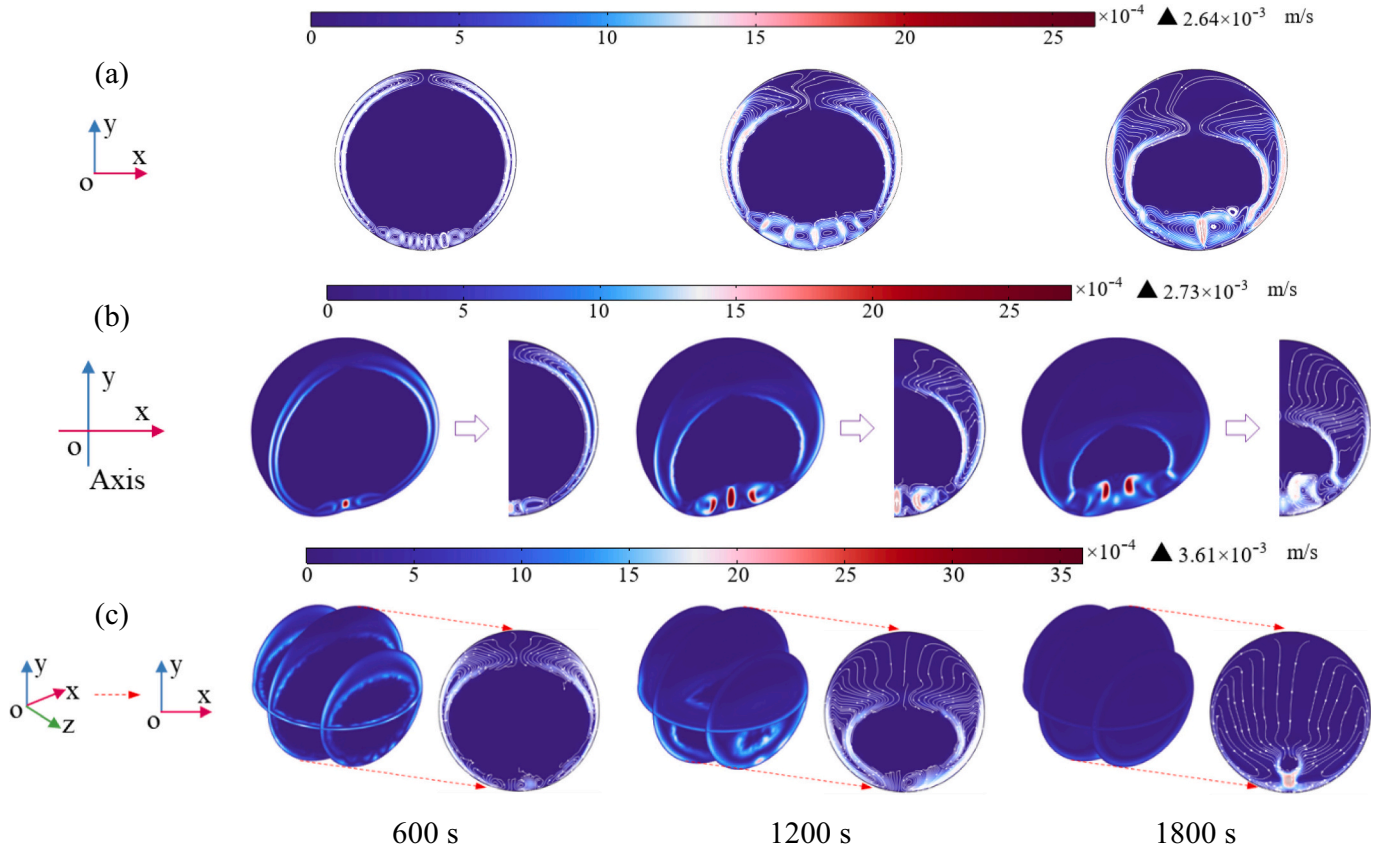


Fig. 10. The evolution process of flow fields over time when $R = 50$ mm. (a) 2D circular model (b) 2D axisymmetric model (c) 3D spherical model.

complete melting of PCM within the encapsulation structure. However, the significance level of influence varies among these factors.

Fig. 14 shows the Pareto plot of the standardized effects of these four factors, where 3.18 is the F-value corresponding to $\alpha = 0.05$. The degree of deviation from the F-value is directly proportional to the significance of the factor. It can be observed that the significance of factors on the time for complete melting is in the following order: $R > A_{mush} > R \times A_{mush} > \Delta T$. Figs. 15 and 16 illustrate the graphs of the main and interaction effects for the three factors. In Fig. 15, larger slopes indicate more significant effects. The interaction effect refers to the impact produced by the combination of two factors. In Fig. 16, if the curves are not parallel, it indicates the presence of interaction effects. It can be also observed that the factor with the greatest influence on the time to achieve complete melting is R, followed by A_{mush} in Fig. 15. It can be seen that only the interaction effect between R and A_{mush} has an effect on the melting process in Fig. 16. The interaction effects of R and ΔT, as well as A_{mush} and ΔT, do not appear to be significant.

Based on the above analysis, it can be concluded that the important factors affecting the time required for PCM to reach complete melting are R, A_{mush}, ΔT and ($R \times A_{mush}$). The time required for complete melting of PCM (t_{melt}) within the encapsulation structure is taken as the dependent variable, while the aforementioned four factors are considered as independent variables for conducting a multivariate linear regression analysis. This yields Eq. (14) along with its applicable range, where the coefficient of determination (φ^2) is 0.9836. A higher value of φ^2 approaching 1 indicates a higher degree of fitting accuracy. Therefore, within the range of application shown in Eq. (14), the time required for complete melting of PCM within the 3D spherical encapsulation structure can be accurately predicted. Additionally, if experiments have been conducted on the melting process of PCM in spherical encapsulation structures, numerical simulations are required to obtain additional details regarding the melting process, such as the temporal evolution of temperature and flow fields. Therefore, with the known radius (R) of the encapsulation structure used in the experiment, the measured t_{melt}, and the ΔT of the PCM, the required A_{mush} value can be calculated using Eq. (14) in the numerical simulation. Following this method helps avoid the inaccuracy of randomly selecting A_{mush} values and saves computational resources.

$$\begin{cases} t_{melt} = 32.02R + 3 \times 10^{-6} A_{mush} - 0.4\Delta T + 0.64(R{\cdot}A_{mush}) - 154 \\ 20 \text{ mm} \le R \le 50 \text{ mm} \\ 1 \times 10^5 \text{kg}{\cdot}\text{m}^{-3}{\cdot}\text{s}^{-1} \le A_{mush} \le 1 \times 10^8 \text{ kg}{\cdot}\text{m}^{-3}{\cdot}\text{s}^{-1} \\ 2 \text{ K} \le \Delta T \le 5 \text{ K} \end{cases} \quad (14)$$

4. Conclusion

This study employs a numerical method to investigate the melting process of phase-change spherical capsules in a solar thermal energy storage system. Initially, through analyzing liquid fraction, temperature distribution, flow characteristics, and the time for complete PCM melting, the impact of structural simplification approaches (reducing the 3D spherical encapsulation structure to either a 2D circular or axisymmetric encapsulation structure) on the PCM melting process is studied. Subsequently, based on the 3D spherical encapsulation structure, a three-factor design method is applied to investigate the main and interaction effects of spherical radius (R), mushy zone constant (A_{mush}), and phase-change temperature range (ΔT) on the PCM melting process. Furthermore, a multivariate linear regression is employed to model the complete melting time of PCM. The main conclusions are as follows:

(1) Simplifying the 3D spherical encapsulation structure into a 2D circular encapsulation structure significantly impacts the numerical computation results of the PCM melting process. The simplification of the 3D spherical encapsulation structure into a

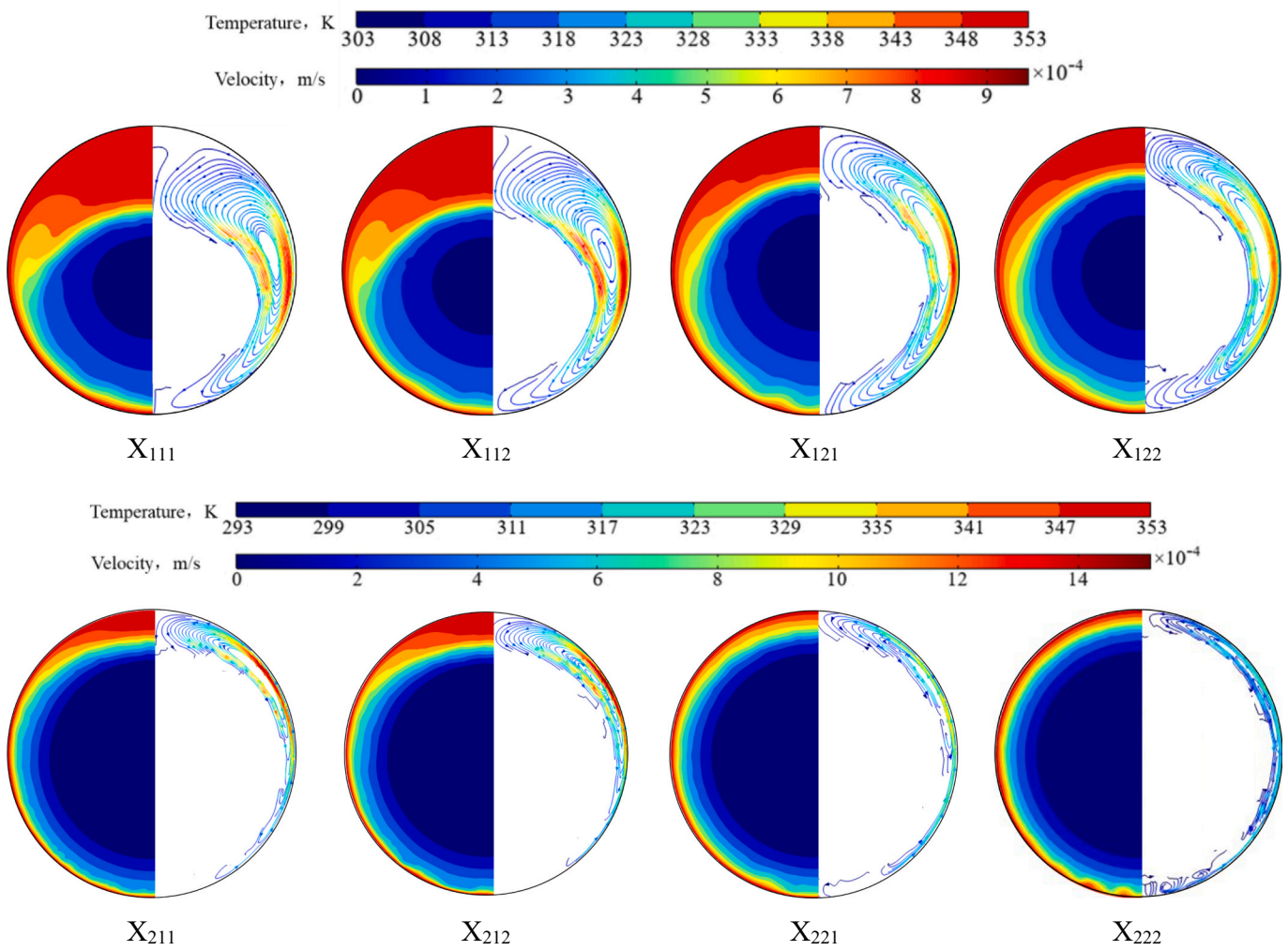


Fig. 11. The temperature and streamline distributions in the *xoy* plane for various conditions at 300 s.

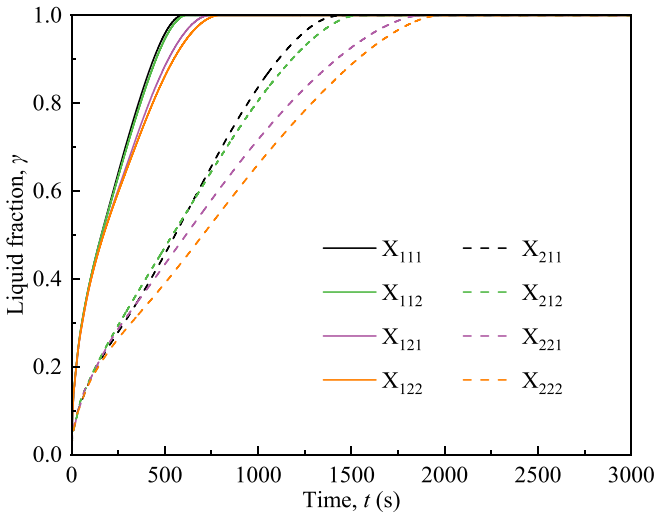


Fig. 12. Variations of liquid fraction with time under different conditions.

2D circular configuration results in a reduction of the heat exchange surface area per unit volume from $3/R$ to $2/R$, leading to a significant decrease in the PCM melting rate. Compared to the 2D circular encapsulation structure, the 3D spherical encapsulation structure illustrates a reduction in the time required for PCM to reach complete melting by 23 % ~ 36 %.

(2) When the radius is small ($R = 10$ mm), the PCM melting process in the 3D spherical encapsulation structure and the 2D axisymmetric encapsulation structure is essentially identical, with only a slight difference of 4.4 s in the time for PCM to achieve complete melting. However, as R increases, the natural convection intensity of liquid PCM and the melting rate of PCM in the 3D spherical encapsulation structure become significantly higher than these in the 2D axisymmetric encapsulation structure. When $R = 50$ mm, the time required for PCM to achieve complete melting in the 3D spherical encapsulation structure is reduced by 509.7 s.

(3) An analysis of the main and interaction effects of the factors is conducted through the three-factor design. The results indicate that the factors R, A_{mush}, ΔT, and $(R \times A_{mush})$ have a significant impact on the PCM melting process. Their significance levels follow the order: $R > A_{mush} > (R \times A_{mush}) > \Delta T$. The computed regression equation obtained through multivariate linear regression analysis of the PCM reaching complete melting time with the four aforementioned factors can accurately predict the time required for complete melting of PCM in the 3D spherical encapsulation structure. Furthermore, when there are changes in the dimensions of the 3D spherical encapsulation structure or the properties of the PCM, it can also provide a basis for the rational selection of A_{mush} in numerical simulations.

Table 5
Time required for complete melting of PCM under different conditions.

Case	X_{111}	X_{112}	X_{121}	X_{122}	X_{211}	X_{212}	X_{221}	X_{222}
Time (s)	589.5	609.9	733.0	787.7	1449.0	1531.2	1872.2	1981.0

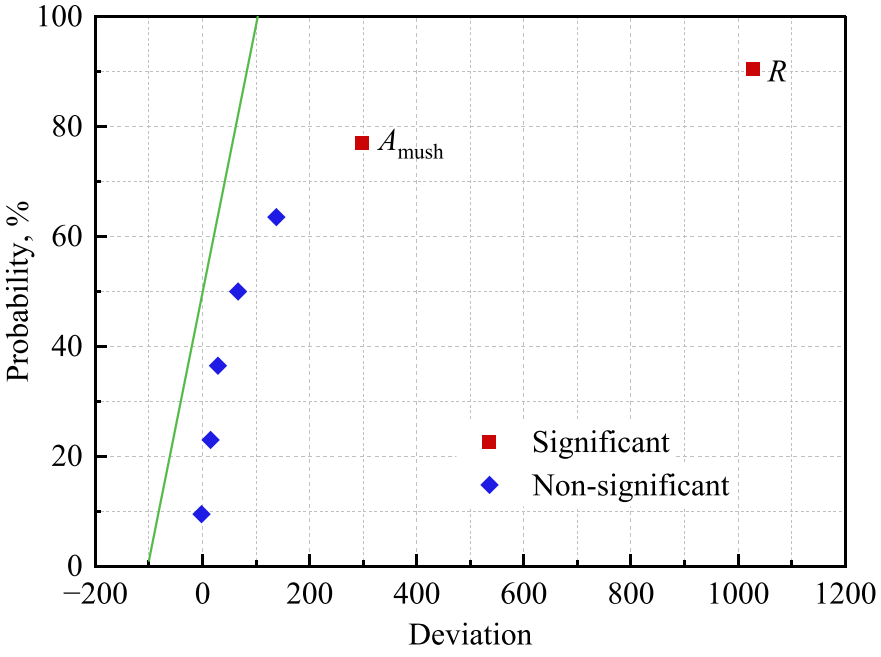


Fig. 13. The normal distribution plots of standardized effects for different factors.

Table 6
Analysis of variance for different factors.

Factor	O	ϕ	F_0	P_0	F (0.05)
R	2,115,213	1	2968.77	0.000	3.18
A_{mush}	178,384	1	250.37	0.001	3.18
ΔT	8831	1	12.39	0.039	3.18
$R \times A_{mush}$	38,088	1	53.46	0.005	3.18
$\underline{E}$	712	3	/	/	3.18

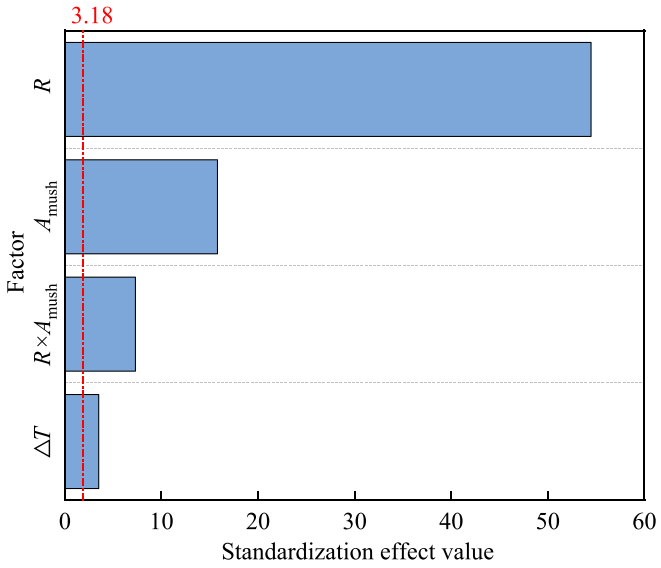


Fig. 14. The Pareto plot of standardization effect for different factors.

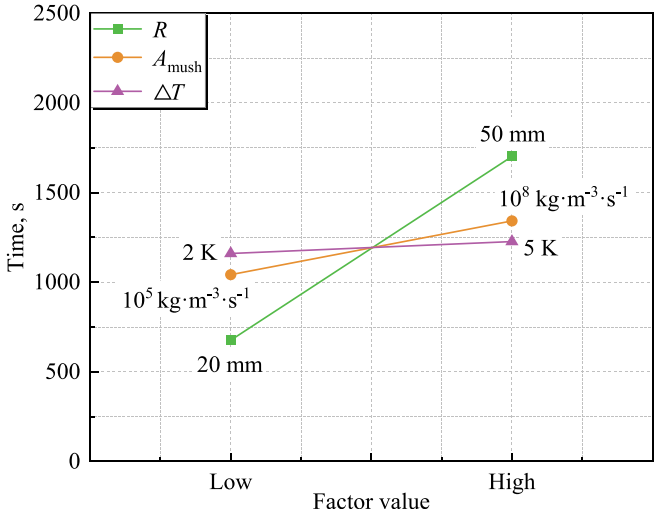
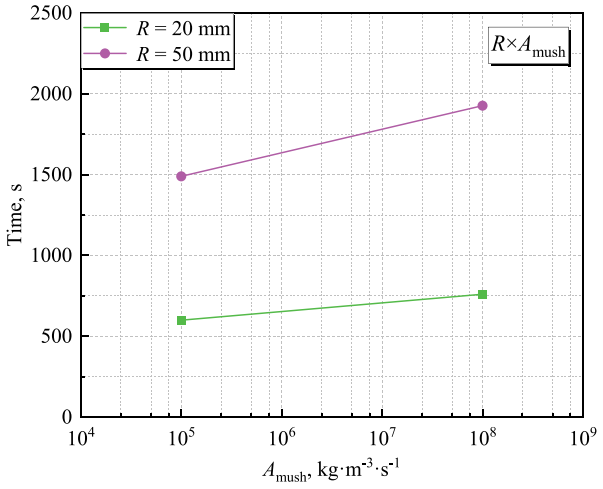


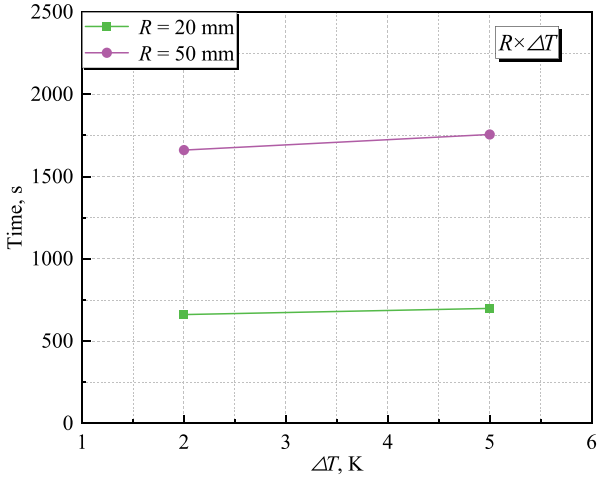
Fig. 15. The main effect of three factors.

Nomenclature

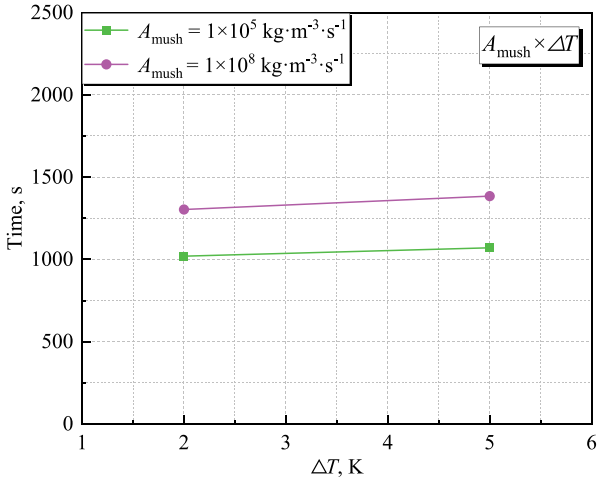
A_{mush}	mushy zone constant, kg·m^{-3}·s^{-1}
$\underline{A_{mush_j}}$	effects of factor A_{mush}
$\left(\underline{A_{mush}\Delta T}\right)_{jk}$	effects of factor ($A_{mush} \times \Delta T$)
a, b, c	number of levels
c_p	specific heat capacity, J•kg^{-1}•K^{-1}
$\underline{E}$	error factor
$\underline{e}_{ijk}$	effect of error factor



(a) The interaction between R and A_{mush}.



(b) The interaction between R and ΔT.



(c) The interaction between A_{mush} and ΔT.

Fig. 16. The interaction effect of three factors.
(a) The interaction between R and A_{mush}.
(b) The interaction between R and ΔT.
(c) The interaction between A_{mush} and ΔT.

F_b	buoyancy, N
G	mean square
H	enthalpy, J•kg^{-1}
ΔH	latent enthalpy, J•kg^{-1}
h	sensible enthalpy, J•kg^{-1}
g	acceleration of gravity, m•s^{-2}
k	thermal conductivity, W•m^{-1}•K^{-1}
L	latent heat, J·kg^{-1}
O	sum of square
R	radius, m
$\underline{R_i}$	effects of factor R_i
$\left(\underline{RA_{mush}}\right)_{ij}$	effects of factor ($R_i \times A_{mushj}$)
$\left(\underline{R\Delta T}\right)_{ik}$	effects of factor ($R_i \times \Delta T_k$)
$\left(\underline{RA_{mush}\Delta T}\right)_{ijk}$	effects of factor ($R_i \times A_{mushj} \times \Delta T_k$)
r	radius, m
$S(T)$	momentum source term
Ste	Stefan number
T	temperature, K
T_m	average phase-change temperature, K
T_{ref}	initial temperature, K
T_w	wall temperature, K
ΔT	phase-change temperature range, K
$\underline{\Delta T_k}$	effects of factor ΔT
t	time, s
V	velocity, m•s^{-1}
x_{ijk}	result obtained in the R_i, A_{mushj}和ΔT_k
x^*	average of different factors
x^{**}	total mean

Greek symbols

α	significance level
β	volume thermal expansion coefficient, K^{-1}
γ	liquid fraction
ε	the constant in Eq. (5)
μ	dynamic viscosity, kg•m^{-1}•s^{-1}
ρ	density, kg•m^{-3}
σ	the variance in Eq. (7)
ϕ	degree of freedom
φ^2	coefficient of determination

Subscripts

i, j, k	the sequence number of levels
l	liquid
s	solid

Abbreviations

PCM	phase change material
STESS	solar thermal energy storage system
2D	two-dimensional
3D	three-dimensional

CRediT authorship contribution statement

Haonan Jia: Conceptualization, Investigation, Methodology, Writing – original draft. **Ziyi Zhou:** Methodology, Writing – review & editing. **Yuhang Tian:** Conceptualization, Writing – review & editing. **Qiuwang Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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